

Emergent Dimensionality and Metastable Oriented Structures in Minimal Discrete Relational Networks

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Abstract

We introduce DRK (Discrete Relational Kernel), a stochastic relational network model based on three local principles:

- (1) persistent relational existence,
- (2) time-asymmetric local reconfiguration,
- (3) adaptive connectivity feedback.

The model generates dynamically evolving networks without predefined geometry, coordinates, or dimensional background.

Numerical simulations show monotonic growth of effective shell dimension with network size across all tested realizations. A local degree-adjustment mechanism (Rule R) prevents connectivity freezing without global information and stabilizes the network near a characteristic average degree.

The framework also produces metastable oriented subgraphs ("nexons") formed by oriented 4-cycles with reinforced local flow structure.

The adaptive rule

```
f = max(1, round(2·D_oriented))
```

frequently produces four-node structures near the transition between one- and two-dimensional growth regimes.

The present work does not claim recovery of general relativity, quantum mechanics, or Standard Model physics.

Instead, it proposes a minimal exploratory framework in which dimensional growth and metastable oriented structures emerge jointly from purely local stochastic rules.

1. Introduction

The question of whether spacetime is fundamentally continuous or discrete has motivated several research programs in quantum gravity. Causal Dynamical Triangulations (CDT) [1] recovers four-dimensional geometry from a path integral over simplicial complexes. Causal set theory [2] derives spacetime from partial orders of discrete events. Wolfram Physics [3] explores hypergraph rewriting systems as models of physical law. In all these approaches, the goal is to show that macroscopic spacetime geometry — including its dimensionality — emerges from simpler discrete structures.

A common feature of these frameworks is the introduction of explicit geometric input: simplices in CDT, a fixed partial-order axiom in causal sets, or specific rewriting rules chosen to match known physics. Here we ask a more minimal question: can spatial dimension emerge from a network that begins with a single node and a single loop, governed only by local rules with no geometric prejudice?

We introduce the Discrete Relational Kernel (DRK) framework, defined by three axioms and one local reconfiguration rule. The framework contains no coordinates, no metric, no pre-assigned dimension, and no global information accessible to individual nodes. We show numerically that:

1. Effective shell dimension $D_{\text{nonoriented}}$ grows monotonically with network size N in all tested realizations.
2. An adaptive connectivity rule based on D_{oriented} outperforms random connectivity, suggesting that dimensional feedback may play a nontrivial structural role.
3. A minimal four-node subgraph — the nexon — is topologically stable under the reconfiguration dynamics.
4. A local degree-adjustment rule homogenizes node connectivity without any global target, eliminating teleological assumptions.

The paper is structured as follows. Section 2 defines the model axioms, the nexon, and Rule R. Section 3 presents numerical results. Section 4 discusses the fixed-point argument for $D_{\text{nonoriented}}$ and D_{oriented} , comparison with related work, and open questions. Section 5 concludes.

2. The Model

2.1 Axiom 0: Existence

The primitive object is a node u carrying a permanent unoriented loop (edge (u,u)). This loop is never destroyed and represents the minimal internal state of a node: its identity. A node without a loop has no relational content; it does not exist in the relational sense.

The permanent unoriented loop is distinct from oriented loops that may emerge through reconfiguration (Section 2.2) and does not contribute to the expansion threshold θ .

The model is a stochastic dynamical system: all edge creation and destruction events are governed by independent Bernoulli trials at each time step. No deterministic trajectory is assumed; results are reported as statistics over independent realizations.

2.2 Axiom 1: Orientational Dynamics

The network has two layers:

Layer 1 — Non-oriented backbone (permanent):

All non-oriented edges, including the permanent loop on each node (Axiom 0), define topological neighborhood. They never decay.

Layer 2 — Oriented dynamic edges:

An oriented flow state may exist on top of an existing non-oriented edge. It carries an orientational weight $w_{or} \geq 1$ representing the intensity of directional flow.

Two constants p_c, p_{d0} in $(0,1]$ with $p_c > p_{d0}$ govern Layer 2:

Creation (probability p_c per node per step):

Node u attempts to create or strengthen an oriented edge:

- Neighbor w exists (non-oriented edge (u,w) exists):
 - No oriented edge yet: create with random orientation (50:50), $w_{or} = 1$.
 - Oriented edge exists: new attempt is random (50:50);
if orientation matches existing $\rightarrow w_{or}(e) += 1$;
if opposite $\rightarrow$ rejected (antiparallel forbidden).
- No neighbor: oriented loop $(u \rightarrow u)$ above the permanent non-oriented loop (Axiom 0). Both directions of $u \rightarrow u$ are equivalent — no orientation conflict. w_{or} increments as above.

Decay (probability $p_d(e)$ per oriented edge per step):

$$p_d(e) = p_{d0} / (1 + \beta \cdot C(e))$$

where $C(e)$ = number of oriented cycles of length ≤ 4 containing e
(computed only from the distance-2 neighborhood of e), β is a free parameter
($\beta = 0.1$ in tests).

```
w_or(e) → w_or(e) - 1.
```

```
When w_or(e) = 0: oriented edge disappears; non-oriented backbone remains.
```

Edges in oriented cycles ($C(e) \geq 1$) decay slower — local cycle-based stabilization.
The inequality $p_c > p_{d0}$ is the sole source of time asymmetry.
No other direction of time is imposed.

2.3 Axiom 2: Expansion

Expansion trigger:

A node u relaxes when its total oriented weight reaches θ :

```
trigger(u) = w_or(loops of u) +  $\sum$  w_or(out-edges of u)  $\geq$  theta
```

The number of
new nodes added per relaxation event and their initial connectivity are
both governed by a single formula:

```
f = max(1, round(2 · D_oriented))
```

where D_{oriented} is the exponentially smoothed D_{shell} of the oriented subgraph
(edges with direction $\neq 0$) over the last T_{avg} expansion events.

Two quantities govern the expansion event:

```
f          = max(1, round(2 · D_oriented))  [inflation: nodes per event]  
k_initial  = 1                             [minimal coupling at birth]
```

f determines how many new nodes are added per relaxation — the inflation rate. It depends on current D_{oriented} and grows with dimension.

Each new node is born with exactly one edge (to its parent) and one permanent unoriented loop

(Axiom 0). Initial connectivity is minimal by design: the node has no information about the global network. The Rule R (Section 2.4)

then builds connectivity dynamically from $k=1$ toward $k_{\text{target}} = \text{round}(2 \cdot D_{\text{nonoriented}})$.

Axiom 2 governs inflation (f): how many nodes
Rule R governs structure ($k \rightarrow k_{\text{target}}$): how many edges

This separation is heuristically motivated: A newly created node does not begin with maximal connectivity.

The adaptive rule for f is currently heuristic and motivated by the coordination number of regular lattices (2D neighbors in d dimensions). Its role is exploratory rather than derived.

2.4 Rule R: Spatial Reconfiguration

Rule R is not an axiom — it is a mechanistic rule ensuring that the non-oriented spatial structure does not freeze at its initial state. It operates independently of Axiom 1 and introduces no new parameters.

Iteration: Rule R iterates over pairs of nodes connected by a non-oriented edge — not over individual nodes. A non-oriented edge (u,v) is a symmetric bond; both nodes participate equally.

Edge creation: For each non-oriented edge (u,v) , with probability p_c , a new non-oriented edge is added to a neighbor of u or v :

```
candidate w: reachable in 2 steps from u or v,  
              not already a direct neighbor of both u and v,  
               $\text{deg}(w) < \text{deg\_local\_avg}(w)$ 
```

Both endpoints must be below their local neighborhood average ($\text{deg}(u) < \text{deg_local_avg}(u)$ AND $\text{deg}(w) < \text{deg_local_avg}(w)$). One candidate is selected uniformly at random.

No edge removal: All non-oriented edges are permanent (Axiom 1 permanence rule). Rule R only adds edges, never removes them.

This rule is conceptually analogous to chemical valence: a bond forms only when both endpoints have available binding capacity, determined entirely by local neighborhood information.

2.5 The Nexon: Elementary Stable Subgraph

A nexon N is a connected subgraph G' of G satisfying:

1. $|V(G')| = 4$ (exactly)

2. G' contains exactly one oriented 4-cycle with $O(e) > O_{\min}$
3. Each node carries exactly one loop
4. $e(G') = 4 > |V(G')| - 1 = 3$ (redundancy satisfied automatically)
5. G' is invariant under the reconfiguration dynamics for at least T_s steps

The type of loop on each node is determined by its role in the oriented cycle:

Role	Degree	Loop type	Orientational role
Source (out=2, in=0)	active	oriented loop (out)	source-like
Sink (out=0, in=2)	passive	oriented loop (in)	sink-like
Relay (out=1, in=1)	neutral	permanent loop only	relay

Three distinct orientations of the 4-cycle yield three chiral classes (R, G, B), whose symmetry group contains Z_3 as a subgroup — a group-theoretic coincidence sharing some algebraic features with Z_3 structures resembling certain finite symmetry structures known from mathematical physics (see Section 4.3). A fourth orientation type (Type 4)

has no natural annihilation partner under the chiral compatibility rules and is a candidate for a long-lived isolated chiral structure.

3. Results

3.1 Emergent Dimensionality

The model maintains two shell dimension estimates, each computed from the shell growth profile $S(k) \sim k^{(D-1)}$ around the topological center of the respective subgraph:

```

D_nonoriented = D_shell of the non-oriented subgraph (direction = 0)
                Used by Rule R for spatial reconfiguration.
D_oriented     = D_shell of the oriented subgraph (direction ≠ 0)
                Used by Axiom 2 for inflation: f = round(2·D_oriented).

```

Both are smoothed over the last $T_{\text{avg}} = 10$ expansion events.

Results in this section report $D_{\text{nonoriented}}$ as the primary measure of emergent spatial dimensionality.

Figure 1 shows $D_{\text{nonoriented}}$ and D_{oriented} as functions of N for a representative realization. Both grow monotonically from 0, with $D_{\text{nonoriented}}$ leading in the early phase and D_{oriented} catching up as oriented connections accumulate.

Table 1. D_{shell} growth with N (seed=42, with Rule R).

N	$D_{\text{nonoriented}}$	D_{oriented}
208	0.00	0.46
544	2.90	1.08
1108	4.01	2.30
2000	4.04	3.25

Both dimensions grow monotonically. D_{oriented} converges toward 3 at $N \approx 2000$, consistent with the fixed-point argument (Section 4.1).

Small- N note: At $N \leq 500$, Rule R appears to reduce $D_{\text{nonoriented}}$ relative to the no-Rule-R baseline. This is an artifact of small network size: at small N , Rule R shortcuts reduce the effective diameter. At larger N (≥ 1000), Rule R is essential for dimensional growth — without it, the network freezes in a star topology with $D_{\text{nonoriented}} \approx 0$.

3.2 Adaptive vs. Random Inflation (Null Model)

The physical model separates inflation (f , governed by Axiom 2) from connectivity ($k = 1$ at birth, built by Rule R). We test two null hypotheses against the adaptive model (10 seeds, 220 steps):

- H0a — No inflation ($f = 1$):** one node per relaxation, $k = 1$.
Tests whether adaptive inflation (growing f with D_{oriented}) contributes at all.
- H0b — Random inflation (f uniform in $\{1, \dots, 6\}$):** random node count, $k = 1$.
Tests whether D_{oriented} -dependent f outperforms arbitrary inflation.

Model	f rule	$D_{\text{nonoriented}}$	Better vs adaptive
Adaptive	$\text{round}(2 \cdot D_{\text{oriented}})$	2.25	—
H0b: random f	uniform $\{1, \dots, 6\}$	2.22	4/10
H0a: no inflation	$f = 1$	2.04	3/10

Key results:

- Adaptive f outperforms $f=1$ ($\Delta = +0.207$, 7/10): inflation driven by $D_oriented$ contributes positively to dimensional growth.
- Adaptive f vs random f is weak ($\Delta = +0.026$, 6/10): the specific adaptive rule $round(2 \cdot D_oriented)$ does not strongly dominate random inflation at current simulation scales.

The second result is reported honestly: the adaptive rule for f has a directional advantage over no inflation, but its superiority over arbitrary inflation is not statistically robust in the present experiments. Larger N and more seeds are needed to resolve this question.

3.3 Nexon Stability

We embed a single nexon in a surrounding network of 20 nodes and run the reconfiguration dynamics for 200 steps (50 independent realizations).

The nexon has two components with distinct stability properties:

Non-oriented backbone (4 permanent edges): survives in 50/50 realizations (100%). The permanence rule of Axiom 1 guarantees this: non-oriented edges are never removed. The four nodes of the nexon remain connected regardless of oriented dynamics.

Oriented 4-cycle (dynamic flow): survives in 35/50 realizations (70%) at $T = 200$ steps, with dynamic decay rate $p_d(e)$ active ($\beta = 0.1$). Without dynamic p_d , survival drops to 23/50 (46%). The reinforcement rule applies the 50:50 filter — a new attempt succeeds only if its random orientation matches the existing one, giving an effective reinforcement rate of $p_c/2$. This reduces the average edge weight to 2.82 (previously 4.61 under deterministic reinforcement), but is physically more transparent. Topological protection ($C(e) \geq 1 \rightarrow p_d(e) < p_d0$) remains active.

This result is structurally meaningful: the nexon is not a permanently frozen structure but a metastable one. The non-oriented backbone remains stable throughout all tested realizations; the oriented flow fluctuates. The 70% survival rate at $T = 200$ reflects the competition between $p_d(e)$ (flow decay with local cycle-based stabilization) and $p_c/2$ (effective reinforcement rate). Longer-lived nexons are expected at higher p_c/p_d0 ratios.

The present framework does not yet contain a rigorous conservation law. Nexons should therefore currently be interpreted as metastable dynamical attractors rather than strictly conserved particles. Identifying possible topological or statistical invariants of the reconfiguration dynamics remains an important open problem.

3.4 Rule R: Preventing Frozen Topology

Every new node is born with $k = 1$ (one non-oriented edge to parent). Without Rule R, nodes remain frozen at degree 1 — the network forms a star topology and $D_{\text{nonoriented}}$ cannot be measured.

Rule R is essential at large N . At small N (≤ 500), Rule R shortcuts reduce the effective diameter and appear to lower $D_{\text{nonoriented}}$. At large N (≥ 1000), this effect is negligible and Rule R enables dimensional growth by homogenizing local degree toward the neighborhood average.

N	Without Rule R	With Rule R
500	$D_{\text{no}} \approx 3.6$ (star artifact)	$D_{\text{no}} \approx 1.8$
1000	frozen (star)	$D_{\text{no}} \approx 3.0$
2000	frozen (star)	$D_{\text{no}} \approx 4.0$

Without Rule R at large N : the network degenerates into a star graph (all nodes connected only to their expansion parent) and dimensional growth stalls. Rule R is structurally necessary to prevent this degenerate topology — nodes without additional connections are relational isolated and cannot participate in nexon formation.

The apparent high D_{no} at small N without Rule R is a transient artifact caused by shell compression around the central hub.

3.5 Parameter Robustness

$D_{\text{nonoriented}}$ grows with the ratio p_c / p_d (monotonically from 1.13 at ratio 1.2 to 1.56 at ratio 2.0), confirming that the time-asymmetry of Axiom 1 directly governs dimensional growth. $D_{\text{nonoriented}}$ decreases with increasing threshold θ (from 1.65 at $\theta=1.5$ to 0.40 at $\theta=5.0$): a higher threshold slows expansion, reducing the number of feedback cycles within a fixed simulation window.

4. Discussion

4.1 Fixed-Point Argument

The rule $f = \text{round}(2 \cdot D_{\text{oriented}})$ is self-consistent at any integer D :

a D -dimensional cubic lattice has coordination number $2D$, so a network with $D_{\text{nonoriented}} = D$ generates nodes with $k_{\text{new}} = 2D$, reinforcing the D -dimensional structure.

Why does the system converge toward $D \approx 2-3$ rather than higher values?

A saturation argument: each new node connects to its parent plus neighbors of the parent. The number of available extra neighbors $\approx \text{deg}(\text{parent}) - 1 \approx k_{\text{new}} - 1$. At $D = 3$ ($k_{\text{new}} = 6$), the requirement of 5 additional connections is marginally satisfiable from the parent's neighborhood. At $D^=4$ the required coordination becomes heuristically difficult to sustain from local neighborhood information alone.*

This argument is heuristic. A rigorous treatment would require showing that (a) the stochastic process for $D_{\text{nonoriented}}$ has a unique stationary distribution, (b) that distribution is concentrated near $D = 3$, and (c) the feedback is stable — small deviations from D are corrected rather than amplified. None of these steps is currently established.

The numerical evidence (monotone growth in 20/20 realizations, low variance) is consistent with convergence but does not constitute a proof.

A formal proof of convergence to a specific fixed point remains an open problem.

4.2 Nexon Threshold and Emergent Structure

The formula $f = \max(1, \text{round}(2 \cdot D_{\text{oriented}}))$ produces $f = 4$ at $D_{\text{oriented}} = 2$.

This is not imposed: it follows from the formula evaluated at the observed transition region between one- and two-dimensional growth regimes. The four-node nexon thus emerges as the characteristic expansion unit at the dimensional transition, without being postulated.

4.2b Network Laplacians

Two distinct Laplacian operators arise naturally in DRK:

L_{un} — symmetric Laplacian of the non-oriented backbone.
Describes effective connectivity geometry.

L_{or} — directed weighted operator of oriented flow states.
Describes propagation and orientational dynamics.

Working hypothesis: effective geometric behavior is primarily associated with L_{un} ; propagation and orientational dynamics with L_{or} . Strong orientational organization may modify the effective network geometry through long-term dynamical feedback.

4.3 Relation to Known Symmetries

The oriented 4-cycle admits several discrete chiral classes under cyclic orientation patterns.

These classes exhibit nontrivial permutation structure and admit a natural mapping to finite permutation groups related to S_4 .

Certain algebraic features — such as Z_3 -like periodicity and three-dimensional irreducible representations — resemble structures that also appear in other areas of mathematical physics.

No physical identification with $SU(3)$, color charge, quark structure, or Standard Model symmetries is claimed.

At present, this observation should be interpreted only as a structural algebraic analogy.

4.4 Comparison with Related Frameworks

CDT recovers a four-dimensional de Sitter spacetime from path integrals over causal triangulations [1]. DRK differs in that dimension is not a target of the path integral but an emergent property of a feedback rule. No sum over histories is performed.

Causal set theory posits discrete events partially ordered by causality [2]. DRK shares the discrete relational ontology but does not pre-impose a causal partial order; the arrow of time emerges from $p_c > p_d$.

Wolfram Physics [3] generates spacetime from hypergraph rewriting. DRK is closer in spirit but uses a simpler primitive (node + loop) and derives its rewriting rule from an explicit local feedback principle (dimensional feedback) rather than by search.

4.5 Limitations and Open Questions

The present framework has several important limitations.

1. No analytical proof of convergence

The fixed-point argument for $D_{\text{nonoriented}} \rightarrow 3$ remains heuristic. A rigorous treatment would require analysis of the stochastic process governing shell-dimension evolution and proof of stability of the adaptive feedback rule.

2. No continuum limit

The relation between the discrete network and a smooth manifold is not established. Preliminary experiments using graph Laplacian operators suggest that large-scale effective geometric behavior may emerge from the network dynamics, but no metric reconstruction or continuum derivation currently exists.

3. No Lorentz invariance

The model does not possess explicit Lorentz symmetry. Whether Lorentz-like behavior can emerge statistically in a continuum limit remains unknown.

4. No quantum dynamics

The current framework is entirely classical and stochastic. No Hilbert space, unitary evolution, or quantum field structure is included.

5. No conservation law

The framework currently lacks a rigorous invariant or conservation law. Nexons should therefore be interpreted as metastable dynamical structures rather than fully conserved particles.

These limitations are characteristic of exploratory discrete relational models and do not undermine the primary numerical results regarding dimensional growth and metastable oriented substructures.

5. Conclusion

We introduced DRK, a discrete relational framework based on persistent existence, local stochastic reconfiguration, and adaptive dimensional feedback.

The framework should currently be interpreted primarily as a stochastic graph-dynamics model with emergent connectivity structure.

Numerical simulations demonstrate that local graph dynamics can generate nontrivial shell-dimensional growth without predefined geometry or global coordination.

The numerical results reported here are exploratory and intended primarily to identify qualitative dynamical behavior.

A local degree-adjustment mechanism (Rule R) prevents connectivity freezing and stabilizes the network near a characteristic average degree using only local information.

The framework also produces metastable oriented subgraphs ("nexons") consisting of oriented 4-cycles with reinforced local flow structure. The backbone of these structures survives in all tested realizations, while the oriented flow itself remains metastable.

The adaptive rule

```
f = max(1, round(2·D_oriented))
```

naturally produces four-node structures near $D_{\text{oriented}} \approx 2$, suggesting a possible structural transition associated with the emergence of stable local organization.

The primary open question is whether $D_{\text{nonoriented}}$ and D_{oriented} converge toward stable fixed points for large network size and whether such convergence can be established analytically.

Future work includes:

Future work will investigate spectral properties of the emergent network geometry, including graph Laplacian spectra and algebraic connectivity.

- rigorous convergence analysis,
- spectral analysis of network Laplacians,
- study of metastable flow structures,
- investigation of continuum limits,
- and analysis of possible emergent geometric behavior.

The present work should be interpreted as an exploratory study of local stochastic relational dynamics rather than a complete physical theory of spacetime or matter.

Reproducibility

All simulations were implemented in Python 3 using standard library modules (`random` , `numpy`). No external simulation framework was used.

These values were chosen empirically as convenient stable operating points. No fundamental significance is currently attributed to them.

Key reproducibility parameters:

- **Parameters:** $P_C = 0.1618$, $P_D0 = 0.1$, $\beta = 0.1$ (coherence parameter, see Section 2.2), $\text{THETA} = 2.718$
- **RNG seeds:** integer seeds in range 42–61 (20 seeds for reproducibility tests, 42–51 for null model and parameter sweeps). Each seed fully determines the stochastic trajectory.
- **Simulation steps:** 250–350 steps per realization (specified per experiment).
- **D_nonoriented/D_oriented computation:** exponential window of $T_avg = 10$ expansion events;
D_shell computed via log-log linear regression on shell profile $S(k)$ for k in $[1, \text{floor}(0.8 \cdot \text{diameter})]$, minimum 3 data points required.
- **Averaging procedure:** all reported means and standard deviations are computed over independent realizations with distinct seeds; no within-run averaging is performed.
- **Implementation language:** Python 3.11. BFS-based shell computation; no compiled extensions.

Source code is available at [repository URL].

Computational Complexity

The dominant cost per simulation step is D_shell computation, which requires BFS from the topological center of the network.

Operation	Complexity	Notes
BFS from one node	$O(N + E)$	N nodes, E edges
Finding topological center	$O(N \cdot (N + E))$	BFS from all nodes
D_shell regression	$O(\text{diameter})$	negligible
Rule R (per step)	$O(k^2)$ per sampled node	$k = \text{avg degree}$

Operation	Complexity	Notes
Full step	$O(N^2 + N \cdot k^2)$	center BFS dominates

For the network sizes used in this work ($N \leq 600$, avg degree ≤ 8), wall-clock time per realization is under 60 seconds on a standard laptop. The $O(N^2)$ scaling of center-finding limits practical simulations to $N \lesssim 2000$ without optimization. For larger N , approximate center estimation (random BFS sampling) would reduce this to $O(N \log N)$.

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